

Technical Blueprint: Autonomous Institutional Quantitative Trading System

This document establishes the architectural requirements and mathematical protocols for an institutional-grade autonomous trading system. The framework is derived from rigorous empirical market microstructure foundations, prioritizing the isolation of the efficient price from transient noise and the strategic management of asymmetric information.

1. Event-Driven OOP System Architecture

The system must operate within a discrete-event framework, rejecting the flawed assumptions of continuous wall-clock time aggregation.

1.1 DataHandler Component: Point Process Logic

The DataHandler is strictly required to process market activity as **point processes**—discrete events occurring in continuous time.

- **Elimination of Simultaneity Bias:** Standard wall-clock aggregation (e.g., 1-minute bars) is prohibited as it induces simultaneity, masking the causal lead-lag relationships essential for high-frequency attribution. The system must utilize time-stamped data (second-level precision) to establish post hoc causality.
- **Event-Time Indexing:** All data must be indexed by **Event Time** (trades/quote updates). This accounts for varying information intensity and ensures the system remains responsive to "bursty" volatility regimes where wall-clock intervals become informationally hollow.

1.2 Strategy and Portfolio Components: Martingale Decomposition

The Strategy module must isolate the **Efficient Price (m_t)** from observed transaction prices (p_t).

- **Structural Decomposition:** Following Hasbrouck (Chapter 7), transaction prices must be treated as a mixture of the random walk (efficient price) and stationary noise (microstructure effects): $p_t = m_t + s_t$.
- **Martingale Protocol:** The core logic assumes the efficient price component follows the Martingale property: $E_{t+1}\{x_t, \dots\} = x_t$. Increments in m_t are treated as unforecastable, representing the true valuation process in a frictionless environment.

1.3 ExecutionHandler Component: Market Mechanisms

The ExecutionHandler must differentiate between **Dealer/Quote-Driven** and **Double Auction/Order-Driven** environments.

- **Calibration Constant:** The handler must utilize the \$0.0079\$ /share average price-to-quote distance as a calibration constant. This adjusts for "dark" liquidity and price improvement not visible on the SIP (Securities Information Processor).
- **Effective Spread Logic:** Implementation must prioritize the **Effective Spread** calculation to account for the disparity between visible quotes and actual execution prices.

2. The ATLAS Multi-Agent LLM Information Framework

The ATLAS framework utilizes multi-agent logic to quantify the informational environment through the lens of asymmetric trade models.

2.1 Asymmetric Information: Quote-Adjustment Logic

Drawing from the **Sequential Trade Model**, the system must detect the presence of **Informed Traders** (μ).

- **Conditional Expectations:** Agents must recalculate Bid (B) and Ask (A) quotes as conditional expectations of value (V) following every trade sign (q_t):
- $A = E[V|Buy]$
- $B = E[V|Sell]$
- **Zero-Profit Condition:** The system maintains logical bounds by ensuring that quotes reflect the expectation that a trade sign carries private information, adjusting beliefs dynamically based on order flow imbalance.

2.2 Strategic Trade and Information Aggregation

Utilizing the **Kyle Strategic Trade Model**, ATLAS agents interpret the impounding of private information.

- **Strategic Distribution:** If the system is identified as an "Informed Trader," it must distribute orders over N auctions to minimize the **Price Impact Coefficient** (λ).
- **Price Informativeness** (Σ_k): The system must track Σ_k to determine when private information has been fully impounded. It is a baseline assumption that approximately half of private information is impounded into the price per auction.

2.3 The Self-Improvement Loop: Wold Theorem Stability

The system updates its information set via the **Innovation** (ϵ_t) defined by the Wold Theorem.

- **Invertible Solution:** To ensure the internal market model does not diverge from reality, the system must strictly adhere to the **Invertible Moving Average (MA)** representation. Non-invertible solutions are rejected as they fail to provide a stable basis for the recursion of the econometrician's information set.

3. Implementation Logic for Walk-Forward Analysis and Risk

3.1 Rolling Time-Series and Stationarity Protocols

- **Forbidden Price Levels:** The use of raw price levels (p_t) in rolling splits is **strictly forbidden**. Because $\text{Var}(p_t)$ increases with t , rolling variance estimates on levels are statistically meaningless.
- **Differencing Requirement:** All analysis must be performed on **price changes** (Δp_t) to ensure covariance stationarity and ergodicity.

3.2 Data Sourcing and Cleansing

- **Institutional Feature Weighting:** Due to the "Old vs. New" paradox and the rapid obsolescence of market institutions, the system must apply a higher weight to recent data.
- **Microstructure Samples:** Protocols must account for the high statistical significance but low economic significance inherent in samples covering many observations over short calendar intervals.

3.3 Transaction Cost Analysis (TCA)

- **Roll Model Calibration:** Fixed costs are estimated as $c = \sqrt{-\gamma_1}$, where γ_1 is the first-order autocovariance of price changes.
- **Fallback Fail-Safe:** If γ_1 is positive in the sample, the system must trigger a fallback logic, as the Roll Model estimate becomes imaginary.
- **Adverse Selection:** The **Generalized Roll Model** must be used to isolate λ , estimating the permanent impact of trades as a proxy for information asymmetry.

4. Mathematical Evaluation and Performance Metrics

4.1 Core Statistical Metrics

Metric, Formula / Formula Basis, Theoretical Authority

Roll Spread Estimate, $2c = 2\sqrt{-\gamma_1}$, Fixed Costs (Ch. 3)

Random Walk Variance, $\sigma_w^2 = \gamma_0 + 2\gamma_1$, Variance of Efficient Price (Ch. 7)

Model Efficiency, σ_s^2 (Pricing Error Variance), Invertible Solution (Ch. 7/11)

Price Impact (λ), $\Delta p_t = \lambda q_t + u_t$, Adverse Selection (Ch. 6)

4.2 Advanced Metrics and Bounded Scores

- **ATLAS Bounded ROI Score:** Calculated as $E\pi = \frac{\sqrt{\sigma_u^2}}{2\sqrt{\sigma_0}}(v - p_0)^2$. The variance of noise trading (σ_u^2) is explicitly treated as the "camouflage" for the system's informed trades.
- **Extreme-Value Precision:** The system must recognize that finite moments for returns exist only up to **order 3**, while volume moments exist only up to **order 1.5**. Standard variance-based metrics for volume are prohibited; power-law scaling metrics must be used.

5. Dynamic Position Sizing and Infrastructure

5.1 Inventory Control and Moving Barriers

Implementation follows the **Amihud-Mendelson Active Inventory Control** model.

- **Monotone Functions:** Bid and Ask quotes must be monotone decreasing functions of the current inventory level (I_t).

- **Moving Barrier Ruin:** Position sizing must account for "Dealer Ruin" caused by **Moving Barriers** (sudden price movements triggering financing defaults) rather than inventory hitting fixed limits.

5.2 Volatility Targeting

- **Uncentered Volatility:** To reduce measurement error in short-horizon microstructure samples, volatility estimates must be **uncentered**. The mean return (μ) must be set to zero for all short-horizon targets to improve estimation precision.

5.3 Infrastructure and LOB Connectivity

- **Limit Order Book (LOB) Logic:** The system shall act as a "Liquidity Provider" via limit orders.
- **Trade Impact Monitoring:** The system must continuously monitor the "Trade Impact Coefficient" to prevent being "bagged" by better-informed participants in continuous electronic markets.

6. Summary of Theoretical Foundations

The system's operational integrity is anchored by two non-negotiable principles:

1. **Martingale Property of Security Prices:** Observed prices consist of the random-walk efficient price and stationary noise. All strategy outputs must isolate the Martingale component for information attribution.
2. **Wold Theorem and Invertibility:** To maintain system stability and linear determinism, all stationary processes must be represented as **invertible moving averages** of innovations (ϵ_t). This ensures the internal model accurately updates following every market event without divergence.